FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
PART TEST – II
PAPER –1
TEST DATE: 11-12-2022

ANSWERS, HINTS & SOLUTIONS

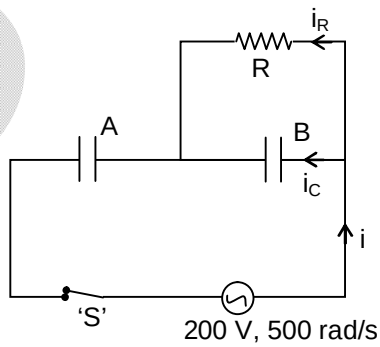
Physics

PART – I

Section – A

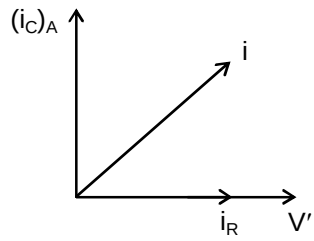
1. A, C, D

Sol. $x_C = \dfrac{1}{\omega_C} = 2000\Omega$ …(i)



$i^2 = (i_C)_A^2 + (i_R)^2$

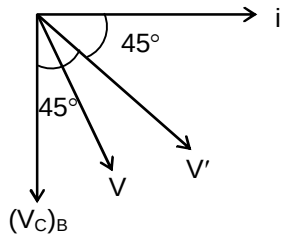
$z' = 1000\sqrt{2}$ …(ii)



$V^2 = (V_C)_B^2 + (V')^2 + 2V_C \cdot V' \cos 45°$

$z = \sqrt{10} \cdot (10)^3$ …(iii)

$i = \dfrac{200}{\sqrt{10} \times 10^3} = \dfrac{1}{5\sqrt{10}}$



$i_R = i\cos 45°$

$(i_C)_A = i\cos 45°$

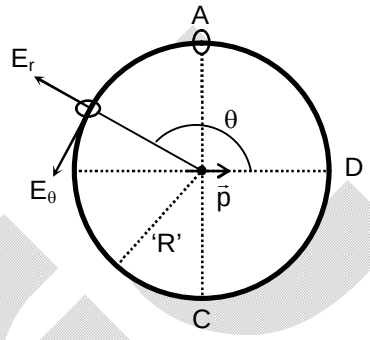
$$\tan\alpha = \frac{V'\sin 45°}{(V_C)_B + V'\cos 45°} = \frac{1}{3} \qquad \ldots(iv)$$

$$\text{Power} = i^2R = \left(\frac{1}{5\sqrt{20}}\right)^2 \cdot 2000 = 4 \text{ W}$$

2. A, C, D

Sol. $E_r = \dfrac{2kp\cos\theta}{r^3}$

$E_\theta = \dfrac{kp\sin\theta}{r^3}$



From conservation of energy at A and E.

$$0 + 0 = \frac{1}{2}mv^2 + \frac{(kp\cos\theta)q}{r^2}$$

$$v = \sqrt{\frac{-2kp(\cos\theta)q}{mr^2}} = \sqrt{\frac{-p(\cos\theta)q}{2\pi\varepsilon_0 mr^2}}$$

3. A, B, C

Sol. Since piston more very slowly so they are at equilibrium at each instant

Above separately wall $P_1 = P_0 + \dfrac{mg}{A}$

Below separately wall $P_2 = P_0 - \dfrac{mg}{A}$

$\Delta Q = 0$

$W + \Delta U = 0$

$W = -\Delta U = -nC_V\Delta T$

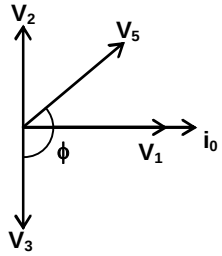
$$P_2V_2 - P_1V_1 = -n\frac{f}{2}R(T_2 - T_1) = \frac{f}{2}\left[nRT_1 - nRT_2\right] = \frac{f}{2}\left[P_1V_1 - P_2V_2\right]$$

$$\left(1+\frac{f}{2}\right)P_2V_2 = \left(1+\frac{f}{2}\right)P_1V_1$$

$P_1V_1 = P_2V_2$, So, $T_1 = T_2$

4. A, C, D

Sol. $e = e_1 + e_2 = 200\sin\left(\omega t + \dfrac{\pi}{2}\right)$



$e_0 = 200$ volt

$v_5 = \dfrac{200}{\sqrt{2}} = 100\sqrt{2}$

$z = \sqrt{(500)^2 + (500)^2} = 500\sqrt{2}$

$i_{rms} = \dfrac{100\sqrt{2}}{500\sqrt{2}} = \dfrac{1}{5}$

$V_1 = 500 \times \dfrac{1}{5} = 100$ volt

$V_2 = 900 \times \dfrac{1}{5} = 180$ volt

$V_3 = 400 \times \frac{1}{5} = 80$ volt

$V_4 = V_2 - V_3 = 100$ volt

$\phi = \frac{\pi}{2} + \tan^{-1}\left(\frac{9}{5}\right)$

5. A, C, D

Sol. $i = \frac{|E_1 + E_2 - E_3 - E_4|}{r_1 + r_2 + r_3 + r_4}$

6. B, C

Sol. $dV = \left(\vec{v} \times \vec{B}\right).d\vec{\ell}$

So potential difference across OA is zero

$\Rightarrow$ Potential difference across OB = $\frac{1}{2}B_0\omega\left(\sqrt{2}OA\right)^2 = \frac{1}{2}(100)(5)2(0.2)^2$ = 20 Volt.

So potential difference across AB is also 20 Volt.

7. A

Sol. $\mu = iA = 0.0550$

$\tau = \vec{\mu} \times \vec{B} = 0.0055$

$|\tau| = I\alpha$

$\alpha = 0.0440$

From conservation of energy

$U_i + K_i = U_f + K_f$

$K_f = -U_f = 0.4400$

8. B

Sol. $\tau = \frac{1}{\frac{6 \times 12}{6+12} + \frac{18 \times 24}{18+24}} = 0.07$

$\Rightarrow$ At steady state

$R_{eq} = \frac{25}{2}$

Current through battery = 2 ampere

Current through inductor = 1/6 ampere

$\Rightarrow$ At one time constant

Current through inductor = $0.63 \times \frac{1}{6} = 0.105$ ampere

From KVL, current through 12Ω = 1.353 ampere

$\Rightarrow$ For part (iv) after switching

$\frac{1}{6}L = i_L(L + 2L)$

$i_L = \frac{1}{18} = 0.056$ ampere

9. C

Sol. $A \rightarrow B$

TP^2 = constant

$PV^{1/3}$ = constant

$$W_{A\to B} = \frac{P_1V_1 - P_2V_2}{x-1} = \frac{nR(T_1 - T_2)}{x-1} = -\frac{9}{2}nRT_0$$

$$\Delta U_{A\to B} = -3nRT_0$$

$$\Delta Q_{A\to B} = -\frac{15}{2}nRT_0$$

$C \to A$

$TP^4 = \text{constant}$

$PV^{1/5} = \text{constant}$

$$W_{C\to A} = \frac{nR(T_1 - T_2)}{x-1} = \frac{15}{4}nRT_0$$

$$\Delta U_{C\to A} = nR(T_A - T_C) > 0$$

$$\Delta Q_{C\to A} > 0$$

$B \to C$

$\Delta U = 0$

$W_{B\to C} > 0$

$\Delta Q_{B\to C} > 0$

10. D

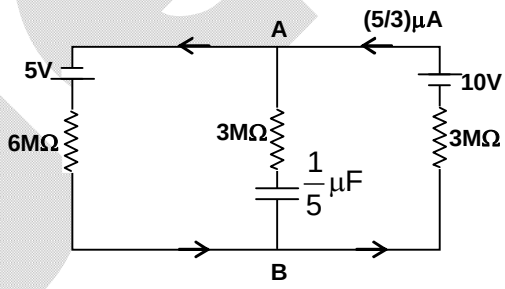
Sol. For τ:

$$R_{eq} = \frac{6\times 3}{6+3} + 3 = 5 \ M\Omega$$

$$\tau = 5 \times \frac{1}{5} = 1 \text{ sec}$$

$\Rightarrow$ For q_{max} at steady state

Current passing through 10 V battery = $\frac{5}{3}\mu A$



$$V_A + 5 - \frac{5}{3}\times 6 = V_B$$

$$V_A - V_B = 5V$$

$q_{max} = 1\mu C$

$\Rightarrow$ At time t = 1sec

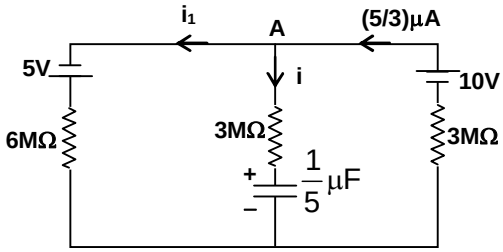
$$q = 1\left(1 - e^{-t}\right) = 0.63\ \mu C$$

$$i = \frac{dq}{dt} = e^{-t} = 0.31\ \mu amp$$

From KVL

$$\frac{0.63}{1/5} + (0.37)3 + 5 - i_1(6) = 0$$

$\mu = 1.54\ \mu A$



Section – B

11. 1.50

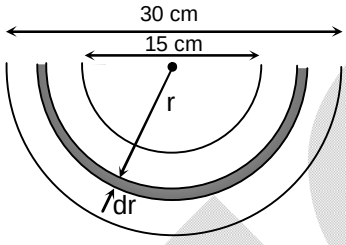
Sol. $$v_{rms}^2 = \frac{\int v^2 dN}{\int dN} = \frac{\int v^2 \frac{dN}{dv} dv}{\int \frac{dN}{dv} dv}$$

$$= \frac{\int_0^3 v^2\left(-\frac{5}{3}v+5\right)dv}{\int_0^3\left(-\frac{5}{3}v+5\right)dv} = 1.50$$

12. 6.93

Sol. $dR = \frac{\rho dr}{\pi r \ell}$

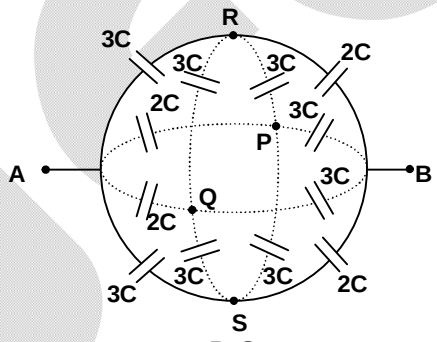
$$R = \frac{\rho}{\pi\ell}\int\frac{dr}{r} = \frac{\rho}{\pi\ell}\ell n2 = 6.93\,\Omega$$



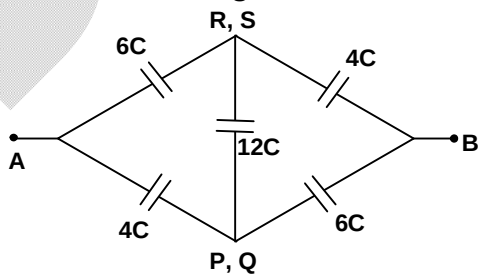
13. 2.10

Sol. P& Q and R & S are equipotential

$$C_{AB} = \frac{2(6C)(4C)+(6C)(12C)+(4C)(12C)}{6C+4C+2(12C)} = \frac{84}{17}C$$



Heat generated = $\frac{1}{2}(C_{AB})V_0^2 = 2.10\ \mu J$



14. 20.30

Sol. Let final pressure on both side of W_2 is P′

$$P'\left(\frac{V_0}{6}\right)^{\gamma} = P_0\left(\frac{5V_0}{6}\right)^{\gamma}$$

$$P' = 5\sqrt{5}P_0$$

Work done by gas = $\frac{5\sqrt{5}P_0\left(\frac{V_0}{6}\right)-P_0\left(\frac{5V_0}{6}\right)}{\gamma-1} = \frac{5P_0V_0}{3}(\sqrt{5}-1)$

$$\Delta U = nC_V\Delta T = n\left(\frac{R}{\gamma-1}\right)\Delta T = 2nR\Delta T$$

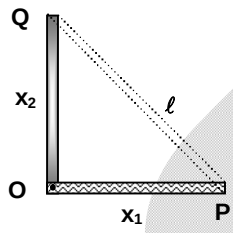
$$= 2\left(P'\frac{5V_0}{6}-P_0\frac{V_0}{6}\right) = 2\left(\frac{5\sqrt{5}P_0 5V_0}{6}-\frac{P_0V_0}{6}\right)$$

$= \frac{P_0V_0}{3}(25\sqrt{5}-1)$

$\Delta Q = \Delta U + W = 2P_0V_9(5\sqrt{5}-1) = 20.3\ P_0V_0$

15. 1.75

Sol. $\ell^2 = x_1^2 + x_2^2$



$2\ell\frac{d\ell}{dt} = 2x_1\frac{dx_1}{dt} + 2x_2\frac{dx_2}{dt}$

$\ell^2\left[\frac{1}{\ell}\frac{d\ell}{dt}\right] = x_1^2\left(\frac{1}{x_1}\frac{dx_1}{dt}\right) + x_2^2\left(\frac{1}{x_2}\frac{dx_2}{dt}\right)$

$\ell^2(\alpha_{PQ}) = x_1^2(\alpha_{PO}) + x_2^2(\alpha_{QO})$

$\alpha_{PQ} = \frac{\alpha_{PO}x_1^2 + \alpha_{QO}x_2^2}{x_1^2 + x_2^2} = 1.7538\times10^{-5}/°C$

16. 1.40

Sol. In first case

$N = mg + iB\ell$

$F_1 = \mu(mg + iB\ell)$

In second case

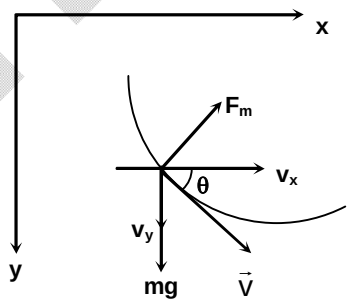
$N = mg - iB\ell$

$F_2 = \mu(mg - iB\ell)$

$\frac{F_1}{F_2} = 1.40$

17. 0.05

Sol. $mg - qBv_X = m\frac{dv_y}{dt}$



$qBv_y = m\frac{dv_x}{dt}$

Differentiate equation (i) with respect to time

$0 - qB\frac{dv_x}{dt} = m\frac{d^2v_y}{dt^2}$

$\frac{d^2v_y}{dt^2} = -\frac{q^2B^2}{m^2}v_y = -\omega^2v_y$

Solution of equation

$v_y = (v_y)_0\sin\omega t$

$\frac{dv_y}{dt} = (v_y)_0\,\omega\cos\omega t$

at t = 0, $\frac{dv_y}{dt} = g$, so $(v_y)_0 = \frac{g}{\omega}$

Now form equation (i)

$v_x = \frac{mg}{qB}(1-\cos\omega t)$

So, $v_y = 0$, at $\omega t = \pi$, $t = \frac{\pi}{\omega}$

$$v_x = \frac{mg}{qB}\left[1-(-1)\right] = \frac{2mg}{qB}$$

From work energy theorem

$$mgy = \frac{1}{2}mv_x^2$$

$$y = \frac{2m^2g}{q^2B^2} = 0.05 \text{ meter}$$

18. 0.08

Sol. In first case

$$R_{eq} = \frac{\frac{5}{2}\times 3}{\frac{5}{2}+3} = \frac{15}{11}\Omega$$

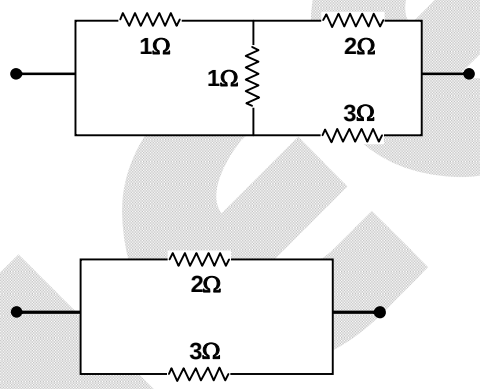
$$i_1 = \frac{V_0}{15/11} = \frac{11}{15}V_0$$

In second case

$$R_{eq} = \frac{2\times 3}{2+3} = \frac{6}{5}\Omega$$

$$i_2 = \frac{V_0}{6/5} = \frac{5}{6}V_0$$

$$\frac{i_1}{i_2} = \frac{22}{25}$$



Chemistry

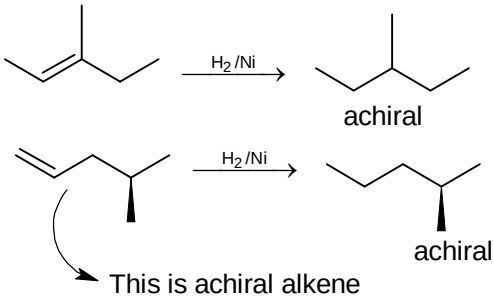
PART – II

Section – A

19. B, C

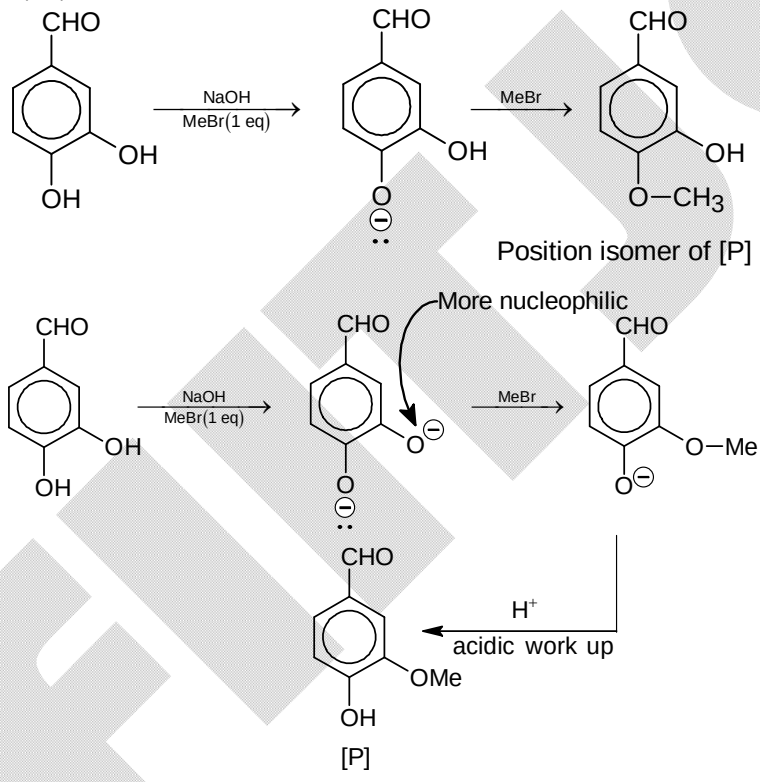
Sol. Alkene [K] has 8α – Hydrogens.

Alkene [L] has 2α – Hydrogens .



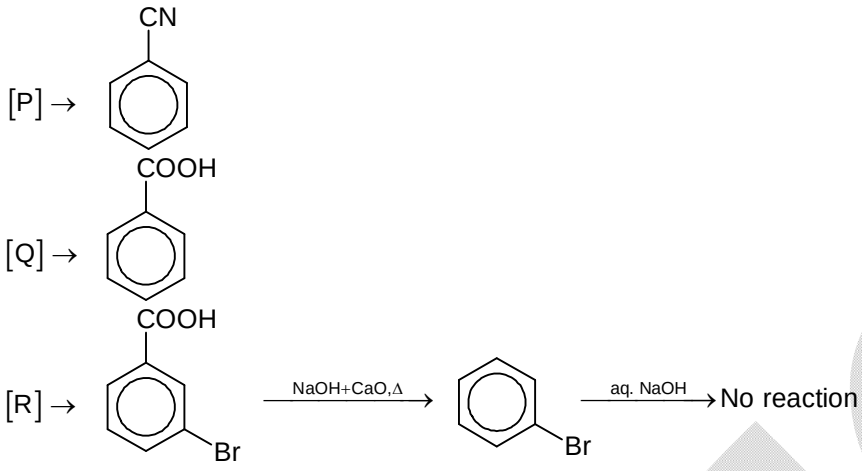
20. B, C, D

Sol.



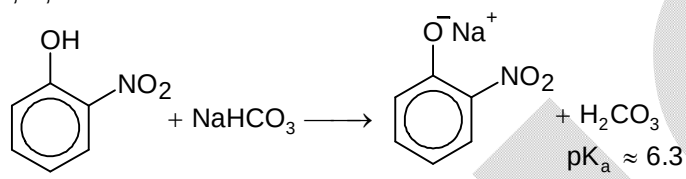
21. B, C, D

Sol.



22. A, B, D

Sol.

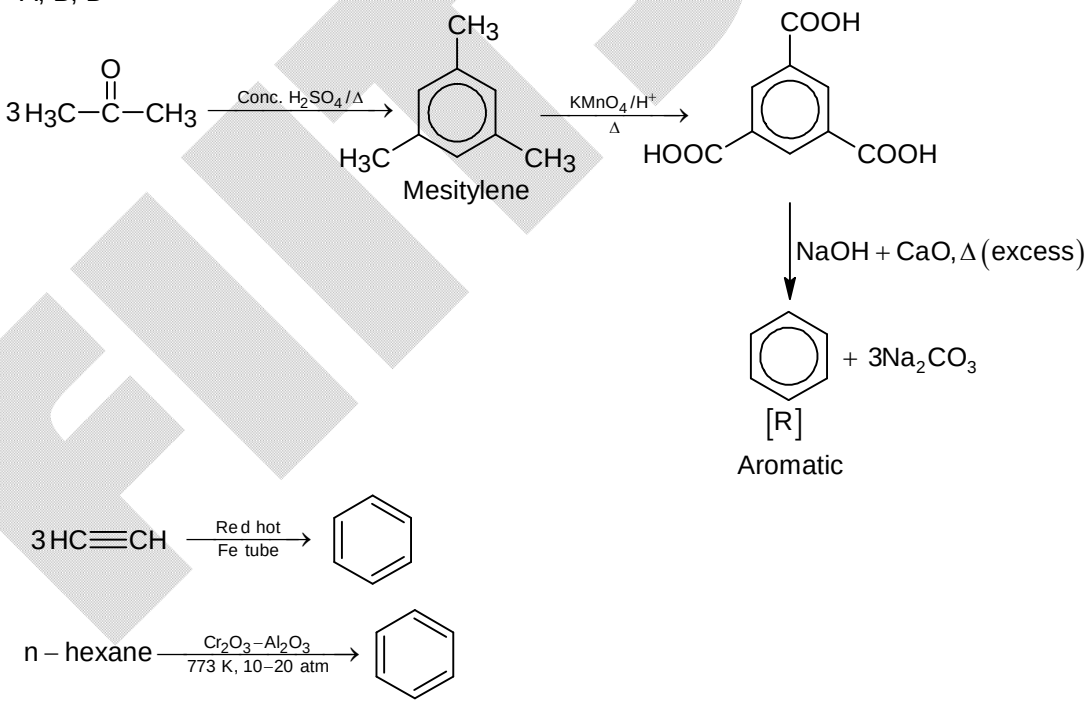


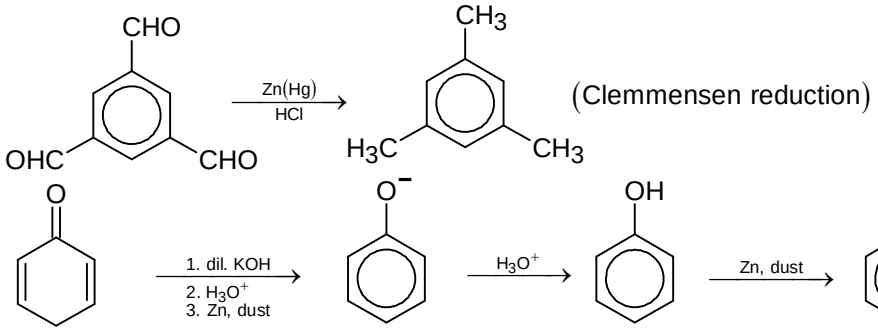
$pK_a \approx 7.2$

As H_2CO_3 is more acidic than o-nitrophenol, so forward reaction is not possible.

23. A, B, D

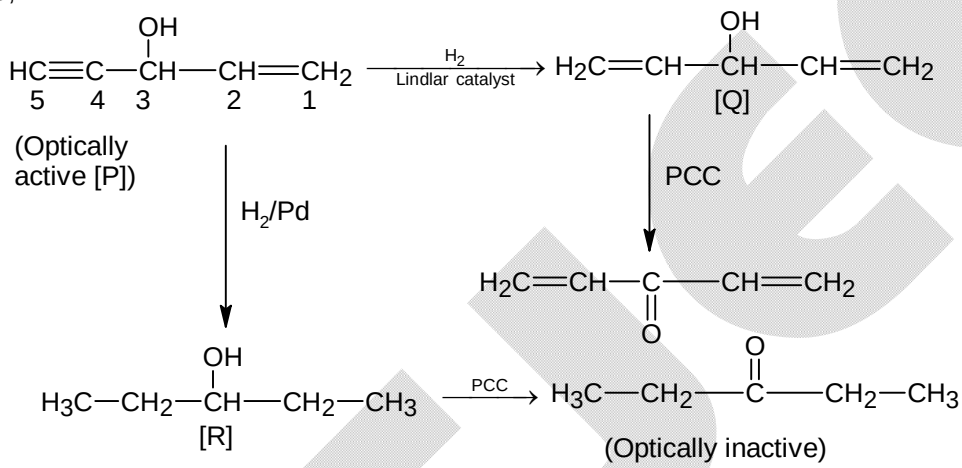
Sol.





24. C, D

Sol.

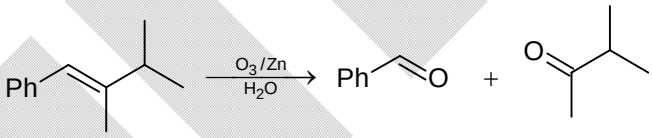


25. B

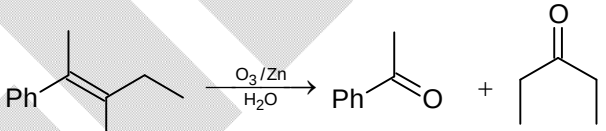
Sol. (I)



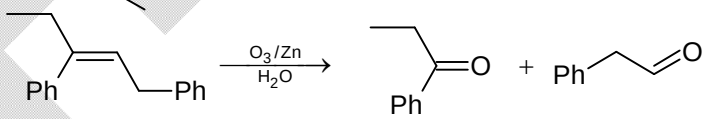
(II)



(III)

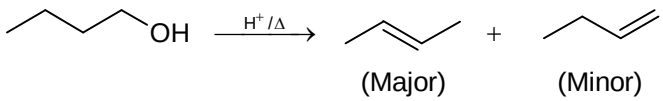


(IV)

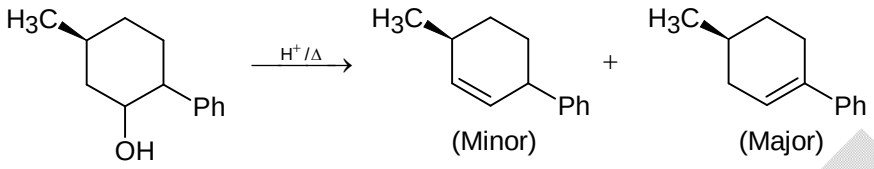


26. D

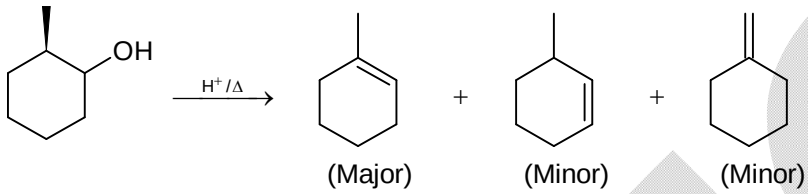
Sol. (I)



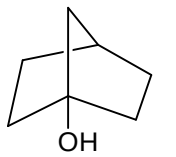
(II)



(III)



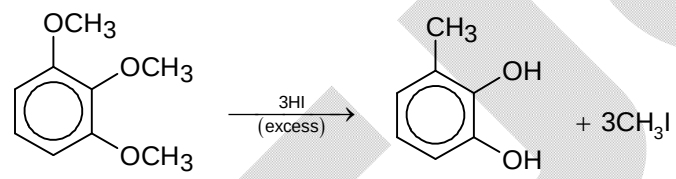
(IV)



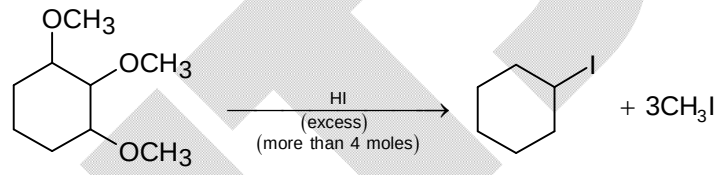
cannot undergo dehydration as bridge head C-atom can never be sp^2-hybridised.

27. D

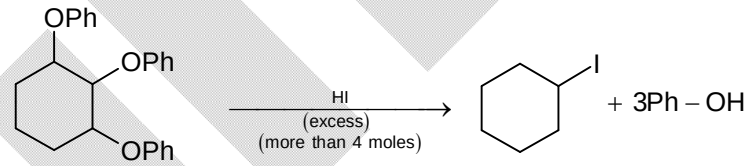
Sol. (I)



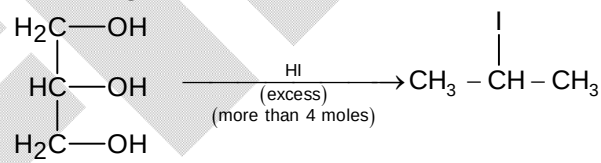
(II)



(III)



(IV)



28. D

Sol. Sucrose $\xrightarrow{\text{Hydrolysis}}$ α – D – glucose + β-D-fructose

Lactose $\xrightarrow{\text{Hydrolysis}}$ β – D – glucose + β-D-galactose

Amylose $\xrightarrow{\text{Hydrolysis}}$ α – D – glucose + α-D-glucose+.....

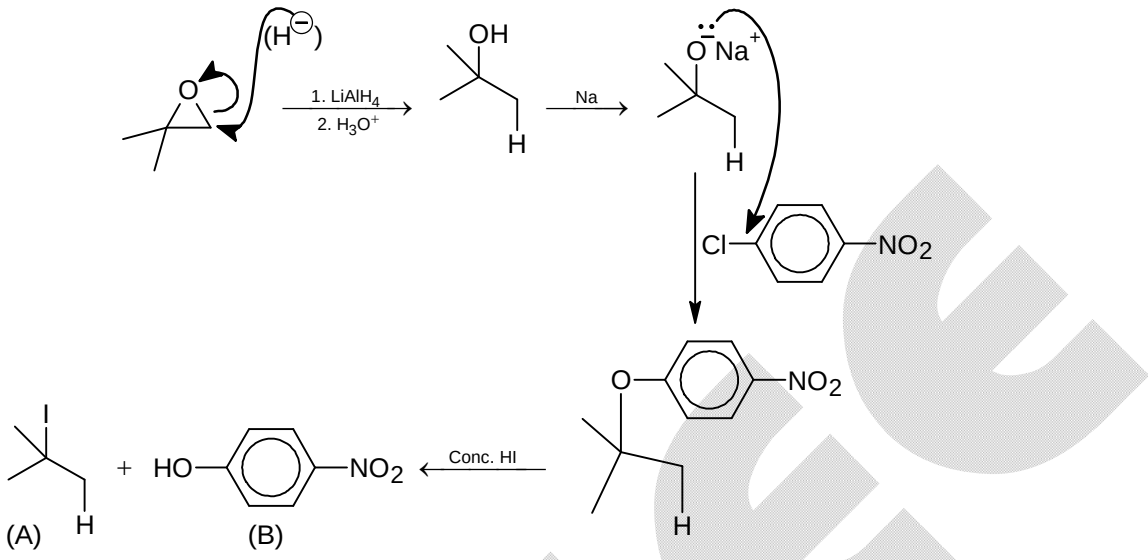
Mannose cannot be hydrolysed.

Lactose and mannose can reduce Fehling's solution due to presence of hemiacetal and aldehyde group.

Section – B

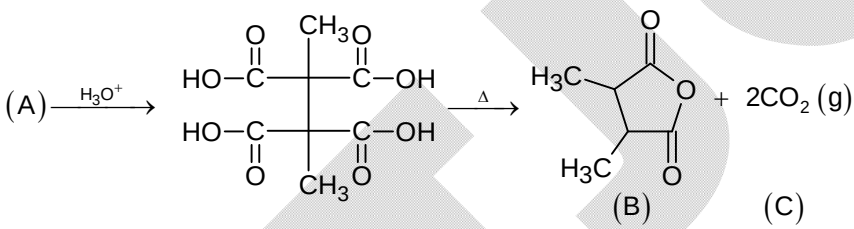
29. 0.50

Sol.



30. 4.40

Sol.



$0.1\,\text{mol A} \rightarrow 0.2\,\text{mol C}$

$n_C = 0.2$

$w_C = 0.2 \times 44\,(g) = 8.8\,(g) = x$

$\frac{x}{2} = 4.40\ \text{grams}.$

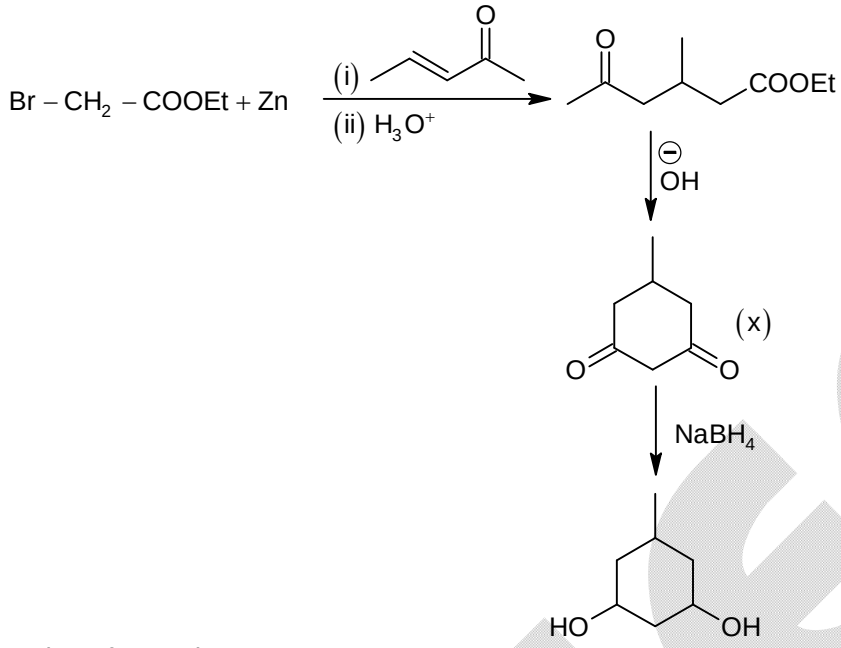
31. 3.50

Sol. 'A' is $CH_3 - CH = C = CH - CH_3$, optically active with minimum weight.

$\therefore x = 5,\ y = 2$.

32. 40.00

Sol.



Number of stereoisomers = 4.

33. 2.00

Sol.

$n = 1$,	$C_2H_3Br_1$	(No stereoisomer)
$n = 2$,	$C_2H_2Br_2$	(Two geometrical isomers)
$n = 3$,	C_3HBr_3	(No stereoisomer)
$n = 4$,	C_2Br_4	(No stereoisomer)

34. 1.20

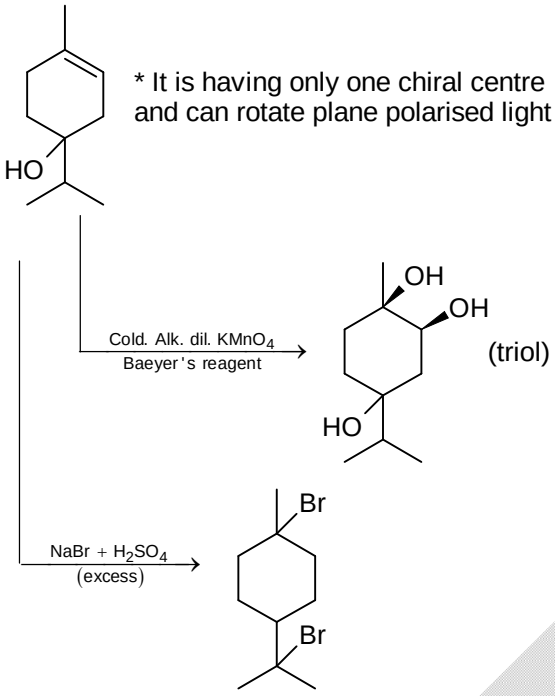
Sol. Co-polymer are Nylon-6,6, Buna-N, Dacron, Buna-S, PHBV. So, X = 5.

Mutarotation is shown by $\alpha - D - G$, $\beta - D - G$, Lactose, $\alpha - D - F$, $\beta - D - F$, Maltose .

So, Y = 6.

35. 1.23

Sol.



36. [18.95 – 18.99]

Sol. Aqueous tension at 287 K = 14 mm of Hg.

Hence, actual pressure = (758 – 14) = 744 mm of Hg.

$$\text{Number of moles of } N_2 = \frac{744 \times 30 \times 10^{-2}}{760 \times 0.0821 \times 287}$$

$$= 1.246 \times 10^{-3} \text{ mol}$$

$$\text{Mass of } N_2 = 1.246 \times 10^{-3} \times 28$$

$$= 34.888 \times 10^{-3} \text{ g}$$

$$\% \text{ mass} = \frac{34.888 \times 10^{-3}}{0.184} \times 100$$

= 18.96%

Mathematics

PART – III

Section – A

37. C, D

Sol. Consider $(x_1, y_1) \equiv (2at, at^2) \Rightarrow$ Equation of tangent $\equiv tx = y + at^2$

$\Rightarrow$ COC w.r.t $(\lambda, t\lambda - at^2)$ is $\lambda(x + ty) - (at^2y + 4a^2) = 0$

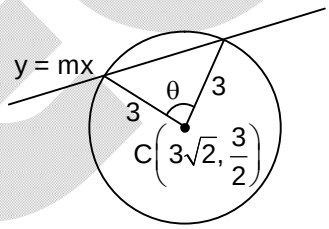
$\Rightarrow (x_2, y_2) \equiv \left(\frac{4a}{t}, -\frac{4a}{t^2}\right)$

38. A, B

Sol. Here, $\cos\theta = \sqrt{\frac{1}{e_1^2} + \frac{1}{e_2^2} - \frac{1}{e_1^2 e_2^2}}$

39. B, C

Sol. $\Rightarrow \frac{\left(\frac{\pi(3)^2}{2\pi}\cdot\theta - \frac{1}{2}(3)^2\sin\theta\right)}{\pi(3)^2 - \left(\frac{\pi(3)^2}{2\pi}\cdot\theta - \frac{1}{2}(3)^2\sin\theta\right)} = \frac{(2\pi - 3\sqrt{3})}{(10\pi + 3\sqrt{3})}$



$\theta = \frac{\pi}{3}$ satisfies this $\Rightarrow m = -\frac{\sqrt{2}}{5}$ or $\sqrt{2}$

40. C, D

Sol. Here, $\frac{(\alpha^2 + \beta^2)^a}{(\alpha\beta)^b}$ is independent of θ if 4a = 5b

41. A, D

Sol. Consider $4x^2 + pxy + 6y^2 = (y - m_1x)(y - m_2x)$, then $4x^2 + qxy - 6y^2 = \left(y + \frac{1}{m_1}x\right)(y - m_2x)$

Comparing, p = –10, q = –2 or p = 10, q = 2

42. A, B

Sol. Given expression = $\frac{(x + y)(x^2 + y^2)}{2}$

43. C

Sol. (I) Here, $x^2 = \sin^2\frac{\pi x}{2} = 1$

(II) $|\tan x| = (4x - \pi)^2 + 16$

(III) $x + y = 0$ or $x = 0 \Rightarrow (x, y) = \left(\pm\frac{1}{2}, \mp\frac{1}{2}\right)$, $(0, \pm1)$, $(\pm1, 0)$

(IV) The given equation becomes, $\frac{(\sin x)\left[(1+\cos x) + (1+\cos x)^2 + 3\cos^2 x\right]}{3} > 0$

$\Rightarrow x \in (0, \pi)$

44. B

Sol. (I) $\cos^{-1} x > 1 > \sin^{-1} x$

(II) $1 > \cos^{-1} x > \sin^{-1} x$

(III) $1 > \sin^{-1} x > \cos^{-1} x$

(IV) $\sin^{-1} x > 1 > \cos^{-1} x$

45. D

Sol. (I) The given parabolas are symmetric about the line y = x

(II) Area of the given triangle = x_0^3

(III) Probability (required) = $\frac{\pi a^2 - \pi ab}{\pi a^2} = \frac{2}{3} \Rightarrow a = 3b$

(IV) Here, roots are $1, \sqrt{2}, -\sqrt{2}$

46. B

Sol. (I) The two given families pass through A(2, 1) and B(19, –4) respectively

$\Rightarrow m_{OA} = \frac{1}{2}$, $m_{OB} = -\frac{4}{19} \Rightarrow$ Slope of $\perp$ lines are –2 and $\frac{19}{4} \Rightarrow \theta = \tan^{-1}\left(\frac{27}{34}\right)$

(II) Given family is always concurrent at $\left(-\frac{1}{6}, -\frac{7}{6}\right)$

(III) $AP^2 + AQ^2 + AR^2 + AS^2 = \frac{7}{4} + \frac{7}{4} + \frac{3}{4} + \frac{27}{4} = 11$

(IV) Given rays are $y = \frac{x}{2}$ and y = –2x ($\perp$) $\Rightarrow$ Required area = $\frac{4\pi - \pi}{4} = \frac{3\pi}{4}$

Section – B

47. 2.00

Sol. If A + B + C = π, then sin 2A + sin 2B + sin 2C = 4 sin A sin B sin C

48. 1011.00

Sol. $\frac{\cot A}{\cot B + \cot C} = \frac{\cos A \cdot \sin B \cdot \sin C}{\sin^2 A}$

49. 4.33

Sol. By the concept of homogenization, $\lambda = \frac{15}{2}$

50. 34937.00

Sol. Here, R(α, β) lies on PQ and midpoint of PR lies on AB

$\Rightarrow \alpha = \frac{23}{7}$, $\beta = \frac{31}{7}$

51. 20.00

Sol. The given ellipse is $(x-3)^2 + \frac{\left(y + \frac{1}{2}\right)^2}{\frac{1}{4}} = 1$

$\Rightarrow x = 3 + \cos\theta, y = -\frac{1}{2} + \frac{1}{2}\sin\theta \Rightarrow 8x - 12y \in [20, 40]$

52. 35.00

Sol. Required number of squares = $5^2 + 3^2 + 1^2 = 35$

53. 0.50

Sol. For $P(t_1)$, $Q(t_2)$, $R(t_3)$. Area of $\Delta PQR = |a^2(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|$

And area of $\Delta ABC = \frac{a^2}{2}\left|(t_1 - t_3)(t_2 - t_1)(t_2 - t_3)\right|$

54. 906.00

Sol. Here, $\cos x = 0$, $\cos 3x = 0$, $\cos 4x = 0$, $\cos 5x = 0$